

ROBERTO YUS

University of Maryland, Baltimore County
1000 Hilltop Circle
Baltimore, MD 21250

✉ ryus@umbc.edu
🏠 <https://robertoyus.com>

EXPERIENCE

University of Maryland, Baltimore County CSEE Department 1000 Hilltop Cir, Baltimore, MD 21250	08/23/2021 – now	Assistant Professor
University of California, Irvine CS Department Irvine, CA 92697	06/01/2016 – 06/01/2021 01/01/2015 – 05/01/2015	Postdoctoral Researcher <i>Supervisor: Sharad Mehrotra</i> Visiting Researcher <i>Host: Sharad Mehrotra</i>
University of Maryland, Baltimore County CSEE Department 1000 Hilltop Cir, Baltimore, MD 21250	06/01/2013 – 05/01/2014	Visiting Researcher <i>Hosts: Anupam Joshi / Tim Finin</i>
University of Zaragoza (Spain) DIIS Department C. de Pedro Cerbuna, 12, 50009 Zaragoza (Spain)	09/01/2010 – 04/01/2016	Graduate Research Assistant <i>Ph.D. Advisor: Eduardo Mena</i>

EDUCATION

Ph.D. in Computer Science <i>Thesis title: "Semantic Management of Location-Based Services in Wireless Environments"</i> <i>Honor/Distinction: Cum Laude</i>	03/17/2016 University of Zaragoza
M.S. in Computer Science <i>Thesis title: "Location-Dependent Query Processing for Live TV Broadcasting of Sport Events"</i>	01/24/2012 University of Zaragoza
B.S. in Computer Science <i>Thesis title: "A Tool to Support Learning Relational Algebra and Query Optimization"</i>	07/16/2010 University of Zaragoza

RESEARCH SUPPORT

External Funding (Total: $\approx$ \$1.7M)

NSF #2542782	\$599,184	07/01/2026 – 06/30/2031
CAREER: Towards a Sensorless Internet of Things through Sensor Data Management Abstraction		
PI: R. Yus		
NSF #2438185	\$969,645	08/15/2025 – 07/31/2028
SFS AI + Cybersecurity: Preparing the Next Generation of Artificial Intelligence and Cybersecurity Professionals		
PI: A. Sherman; Co-PI: R. Yus		
NSF #2432111	\$20,000	07/01/2024 – 12/31/2024
Travel: Fellowships for Students from U.S. Universities to Attend ISWC 2024		
PI: R. Yus ; Co-PI: K. Joshi		
INCS-CoE	\$40,000	06/01/2022 – 12/01/2022
International Digital Trust Framework (Seed Funding)		
PIs: R. Yus , N. Karimi, S. Kai, T. Kondo, M. Sel, N. Panteli, K. Mersinas		
Internal Funding (Total: $\approx$ \$166K)		
UMBC's Strategic Awards for Research Transitions (START)	\$25,000	08/26/2026 – 08/25/2027
Optimizing AI Workflows for Reliable, Low-Cost Agentic Systems		
PI: R. Yus		
UMBC's Hrabowski Innovation Fund	\$24,500	01/26/2026 – 01/26/2028
SPACES: Learning by Making UMBC Smarter - A Reusable Framework for Interdisciplinary, Service-Learning Projects		
PI: R. Yus		
UMBC's Cybersecurity Institute	\$45,500	01/01/2025 – 12/31/2025
Cybersecurity Graduate Fellows program		
PI: R. Yus		
UMBC's COEIT Interdisciplinary Proposal (CIP) Award	\$20,000	09/01/2025 – 8/31/2026
A Multidisciplinary Approach to Developing a Community-driven Framework for Sensorization in Assisted Living Communities		
PIs: R. Yus , T. Reynolds		
UMBC's Center and Institute Departmentally-Engaged Research (CIDER) Grant Program	\$50,000	01/23/2025 – 6/30/2026
Evaluation of standards and methods that claim to prove the authenticity of electronic audio-visual content		
PI: K. Kullman; Co-PIs: E. Golaszewski, R. Yus , A. Sherman		
UMBC	\$1,500	09/01/2022 – 05/01/2023
Supplement for Undergraduate Research Experiences (SURE) award		
PI: R. Yus		

PARTICIPATION IN RESEARCH PROJECTS

Multi-stakeholder Perspectives on a Sensorized Work Environment in Nursing Home Settings, UMB Institute for Clinical & Translational Research. PIs: T. Reynolds and S. Holmes. Participation: May 2023 -- May 2024.
 Developing sensorization plans for nursing homes.

Next-gEnerATion dAta Management to foster suitable Behaviors and the resilience of citizens against modErN Challenges (NEAT-AMBIENCE), Spanish Ministry of Science and Innovation (MICINN),

PID2020-113037RB-I00). PI: S. Ilarri. Participation: Sept 2021 — Aug 2026.

Tackle data management issues to help citizens face modern challenges.

Privacy Cognizant IoT Environment for the Brandeis Program, DARPA (FA8750-16-2-0021). PIs: S. Mehrotra, N. Venkatasubramanian, and A. Kobsa. Participation: May 2016 -- May 2021.

Leading the design/development of TIPPERS, an IoT data management middleware. Focus on design of the interoperability layer and the abstraction from raw sensor data to higher-level semantic inferences.

Creating an Extensible Data Exchange and Analytics Sandbox for Smart Water infrastructures, NSF S&CC (#1952247). PIs: N. Venkatasubramanian, S. Mehrotra, S. Ren, D. Feldman, and R. Eguchi. Participation: Oct 2020 – May 2021

Design and development of an interoperability layer to integrate water related information from diverse sources, including from IoT sensors.

4V: Volume, Velocity, Variety, and Veracity in the Innovative Management of Data, Spanish Ministry of Science and Innovation (CICYT, TIN2013-46238-C4-4- R). PIs: E. Mena and S. Ilarri. Participation: Jan 2014 — Dec 2016.

Adaptation of semantic technologies to support knowledge representation/reasoning/integration in constrained mobile devices.

Semantic Data Management in Wireless Environments, Spanish Ministry of Science and Innovation (CICYT, TIN2010-21387-Co2-02). PI: E. Mena. Participation: Jan 2011 — Dec 2013.

Design of a general and flexible agent-based architecture based on the use of semantic technologies to provide services to mobile users.

AWARDS, RECOGNITION, AND PRESS

Awards:

- CWIT's Outstanding Research Partner Award 2026.
- Best Artifact Award at PerCom 2025.
- 2nd place INCS-CoE Sandpit Challenge on Digital Trust 2022.
- Mark Weiser Best Paper Award at PerCom 2022.
- FPI Fellowship of the Spanish Ministry of Science and Innovation 2011-2015
- Student Travel Award – ISWC (2015,2014), WI-IAT (2015), CollaborateCom (2014)

Press:

- The Responsible AI Digest by SoRAI: *"Teaching AI to Speak 'Privacy': The Need for 'Trustworthy' AI-based Privacy Assistants"*, 2024. [hyperlink]
- Huffington Post: *"Should I Click 'Allow Cookies' On Every Website That Asks?"*, 2022. [hyperlink]
- Publimetro (Mexico): *"Tus dispositivos inteligentes saben más de ti de lo que crees"*, 2022. [hyperlink]
- UMBC CSEE: *"UMBC faculty Karimi and Yus help team win 2nd place in Sandpit Challenge on Digital Trust"*, 2022. [hyperlink]
- Let's Go There w/ Shira & Ryan – Channel Q: *"How your machines are spying on you"*, 2022. [hyperlink]
- The Conversation: *"Smart devices spy on you – 2 computer scientists explain how the Internet of Things can violate your privacy"*, 2022. [hyperlink]
- The Washington Post: *"The Technology 202: Tech to contain coronavirus on college campuses sparks fresh privacy concerns"*, 2020. [hyperlink]
- Defense Visual Information Distribution Service: *"NAVWAR Trident Warrior Team Assesses New Tracking Technology for COVID-19 Mitigation"*, 2020. [hyperlink]
- Health IT Analytics: *"UCI Uses Campus Wi-Fi to Test COVID-19 Contact Tracing App"*, 2020. [hyperlink]

- Military News: *"Navy tests experimental technology at Trident Warrior 2019"*, 2019. [hyperlink]
- UCI CS News: *"IoT Hackathon Apps Coming Soon"*, 2018. [hyperlink]
- UCI CS News: *"ICS launches IoT Hackathon to discover student talent who can develop a UCI 'smart' campus"*, 2017. [hyperlink]
- Nature: *"What could derail the wearables revolution?"*, 2015. [hyperlink]
- Infodocket: *"Interesting Research! FaceBlock App Allows Google Glass Users to Take Privacy-Aware Pictures"*, 2014. [hyperlink]

SERVICE

Program Committee:

- Int. Conf. on Management of Data (SIGMOD) 2026–2027.
- AAAI Conference on Artificial Intelligence (AAAI-27) 2027.
- Int. Conf. on Pervasive Computing and Communications (PerCom) 2025–2027.
- 1st Int. Workshop on Secure and Intelligent Data Spaces (SIDS) @MDM 2026.
- 31st International Conference on Parallel and Distributed Systems (ICPADS) 2025.
- Special Session on Trustworthy and Responsible Data Mining at the 21st International Conference on Advanced Data Mining and Applications (ADMA) 2025.
- Int. Database Engineered Applications Symposium (IDEAS) 2024–2026.
- 13th Int. Conf. on MOBILE Wireless MiddleWARE, Operating Systems, and Applications (MobileWare) 2024.
- 24th Int. Conf. on Distributed Applications and Interoperable Systems (DAIS) 2024.
- 59th IEEE Int. Conf. on Communications (ICC) 2024.
- Int. Conf. on Data Engineering (ICDE) 2023, 2017.
- Symposium On Applied Computing (SAC) - Interoperability Track & IoT and Edge Computing Track - 2023–2026.
- Int. Symposium on Stabilization, Safety, and Security of Distributed Systems (SSS) 2023.
- ACM SIGEnergy Workshop on Fair, Accountable, Transparent, and Ethical (FATE) AI for Smart Environments and Energy Systems (@SenSys/BuildSys) 2021–2022.
- Int. Conf. on Advances in Semantic Processing (SEMAPRO) 2017–2022.
- Int. Conf. on Green Communications, Computing and Technologies (GREEN) 2020–2023.
- Int. Conf. on Big Data, Small Data, Linked Data and Open Data (ALLDATA) 2018.
- Int. Conf. on Advanced Collaborative Networks, Systems and Applications (COLLA) 2016–2019.
- Int. Workshop on Technology Enablers and Innovative Applications for Smart Cities (TESCA) 2019.

Organizing Committee:

- 24th Int. Conf. on Pervasive Computing and Communications (PerCom) 2026 – *Artifact Chair*.
- 23rd Int. Conf. on Pervasive Computing and Communications (PerCom) 2025 – *Local Organizer & Publicity Chair*.
- 26th IEEE International Conference on Mobile Data Management (MDM) 2025 – *Publicity Chair*.
- 22nd Int. Semantic Web Conf. (ISWC) 2024 – *Local Organizer & Student Travel Award Chair*.
- 1st Workshop on Private, Secure, and Trustworthy IoT Data Management (ASTRIDE) at the 39th IEEE International Conference on Data Engineering (ICDE), 2023 – *Organizer*
- 3rd Brain-Computer Interfaces Workshop for Mobile EEG 2023 – *Organizer*
- 7th IEEE Int. Conf. on Smart Computing (SMARTCOMP) 2021 – *Online Session Coordination Chair*.

- 1st Workshop on Mobile Deployment of Semantic Technologies (MoDeST) at the 14th Int. Semantic Web Conf. (ISWC) 2015 – *Organizer*.
- 17th Spanish Congress on Fuzzy Logic and Technologies (ESTYLF) 2014 – *Organizer*.

Journals:

- Frontiers in Big Data 2018–now, *Community Reviewer, Data Mining and Management*.
- Sensors 2020–2026, *Topical Advisory Panel*.
- ACM Transactions on Internet Technology (TOIT) 2026, *Reviewer*.
- IEEE Computer Magazine 2025, *Reviewer*.
- Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 2025, *Reviewer*.
- Expert Systems with Applications 2023, *Reviewer*.
- IEEE Transactions on Artificial Intelligence 2023, *Reviewer*.
- Communications of the ACM 2022, *Reviewer*.
- Pervasive and Mobile Computing 2022, 2019, *Reviewer*.
- Journal of Urban Technology 2021, *Reviewer*.
- IEEE Internet of Things Journal 2019, *Reviewer*.
- Telecommunication Systems 2016, *Reviewer*.
- The Scientific World Journal 2015, *Reviewer*.
- Journal of Internet Services and Applications 2015, *Reviewer*.
- EAI Endorsed Transactions on Ubiquitous Environments 2014, *Reviewer*.
- Distributed and Parallel Databases 2014, *Reviewer*.
- Journal of Systems and Software 2013, *Reviewer*.
- IEEE Multimedia 2013, *Reviewer*.

External Reviewer Conferences:

- Int. Conf. on Extending Database Technology (EDBT) 2016.
- Int. Conf. on Agents and Artificial Intelligence (ICAART) 2016.
- European Conf. on Information Systems (ECIS) 2016.
- Int. Conf. on Very Large Data Bases (VLDB) 2015.
- Extended Semantic Web Conf. (ESWC) 2014.
- AAAI Conf. on Artificial Intelligence (AAAI) 2014.
- Int. Conf. on Ontologies, DataBases, and Applications of Semantics (ODBASE) 2014.

Mentor:

- Center for Women in Technology - UMBC - 2021–now;
- McNair Scholars Program - UMBC - 2023–now;
- Louis Stokes Alliance for Minority Participation - UMBC- 2022–now;
- Meyerhoff Scholars Program - UMBC - 2022–now;
- Institute for African-American Mentoring in Computing Sciences 2023–now;
- Science Champions for Change - UC Irvine - 2021.
- IoT-SITY Research Experiences for Undergraduates (IoT-SITY REU) - UC Irvine - 2017–2020;

- Summer Undergraduate Research Fellowship in the IoT (SURF-IoT) - UC Irvine - 2018–2020;

UMBC:

- Advisory Board – *Member* – Center for Women in Technology - 2025–now;
- CSEE Research Day Committee – *Chair* - 2025–now;
- COEIT Research Day Committee – *Member* - 2025–now;
- CSEE Research Day Committee – *Member* - 2022–2025;
- CSEE Publicity Committee – *Member* - 2023–2024;
- CS MS/PhD Admission Committee – *Member* – CSEE - 2021–now;

Thesis Committee:

- *PhD - Member*: Prajna Bhandary (UMBC, 2026), Yuechun Gu (UMBC, 2025), Elmira Deldari (UMBC, 2025), Ryan Barron (UMBC, 2025), Maksim Eren (UMBC, 2024), Abdulaziz Alshaeri (UMBC, 2024), Anantaa Kotal (UMBC, 2024), Aidin Shiri (UMBC, 2023), Sai Sree Laya Chukkapalli (UMBC, 2022), Naima Khan (UMBC, 2022).
- *MS - Director*: Sai Sragvi Vibhushan Nidamarti (UMBC, 2023), Chetan Mediboyina (UMBC, 2023), Jun Ma (Telecom SudParis, 2022), Zainab Rahal (Telecom SudParis, 2022), Enrique Solano (University of Zaragoza, 2015), Gorka Guerrero (University of Zaragoza, 2012).
- *MS - Member*: Jacob Olinger (Portland State University, 2026), Dylan Conklin (Portland State University, 2025), Aayush Jannumahanti (UMBC, 2024), Raguvir S (UMBC, 2023), Edosa Leta (UMBC, 2023), Maksim Eren (UMBC, 2022).

Other:

- Judge at hackUMBC (2025).
- Panelist at Diversity Panel, Society of Hispanic Professional Engineers, UMBC (2025, 2023).
- Panelist at Research panel, STARS Computing Corps (2024).
- BEST of CWIT Professor Panel (2023).
- INCS-CoE Community Expert (2022 – Present)
- Judge at STEMinate science fair (2022).

TEACHING

CMSC463/663 - Data Privacy Spring 2026, UMBC, 39 students.

CMSC331 - Programming Languages Spring 2026, UMBC, 40 students.

CMSC331 - Programming Languages Fall 2025, UMBC, 38 students.

CMSC463/663 - Data Privacy Spring 2025, UMBC, 40 students.

CMSC331 - Programming Languages Fall 2024, UMBC, 40 students.

CMSC491/691 - Data Privacy Spring 2024, UMBC, 40 students.

CMSC331 - Programming Languages Fall 2023, UMBC, 40 students.

CMSC491/691 - Data Privacy Spring 2023, UMBC, 35 students.

CMSC331 - Programming Languages Fall 2022, UMBC, 40 students.

CMSC491/691 - Data Privacy Spring 2022, UMBC, 40 students.

CMSC331 - Programming Languages Fall 2021, UMBC, 40 students.

RESEARCH ADVISEES

Current Ph.D. advisees:

- *UMBC*: Aamir Hamid (2023–now), Christian Badolato (2023–now).
- *Telecom SudParis*: Jun Ma (2022–now).

Current undergraduates: Nathan Samson (2023–now), Kent Dang (2024–now), Pearce Packman (2025–now), Jalen Brown (2024–now) .

Past Ph.D. advisees (* → Thesis defended):

- *UMBC*: Sanaa Mironov (2023–2025).

Past M.S. advisees (* → Thesis defended):

- *UMBC*: Hemanth Reddy Samidi (2023), Brian Scherzo (2023), Sai Sragvi Vibhushan Nidamarti* (2021–2023), Sarang Patil (2021–2023), Chetan Mediboyina* (2022–2023), Sanaa Mironov (2022), Lasya Chitturi (2022), Dharani Madireddy (2022), Meenakshi Desu (2022), Nikhil Boddu (2022), Akhil Movva (2022).
- *Telecom SudParis*: Jun Ma* (2022), Zainab Rahal* (2021).
- *University of Zaragoza*: Enrique Solano* (2013–2015), Gorka Guerrero* (2012).

Past undergraduates (* → Thesis defended):

- *UMBC*: Gia Oriana Santos (2024–2025), Yazlin Moujalled (2024), Sean Belon (2024), Peter Gatira (2024), Darsh Patel (2024), Yael Chookrun (2024), Shanzé Raza (2023), Brandon Hill (2023), Manav Bhatt (2023), Jahir Bakari (2023), Nikki Cayas (2023), Joshua Cheeks (2022–2023), Luke Zimmermann (2022–2023), Ruhshana Bobojonova (2022–2023) Tartela Tabassum (2022–2023), Aliza Garrison (2022–2023), Emma Williams (2021–2023), Qanita Baqibillah (2023), Diyor Mamarakhimov (2022).
- *UC Irvine*: Aliyah Byon (2020–2021), Vikram Miryala (2020–2021), William Sun (2020), Stuart McClintock (2020), Anna Lubienski (2020), Alpine Tang (2020), Nathan Ma (2020), Yiming Wang (2020), Jason Huang (2020), Leo Peng (2019–2020), Jesse Chong (2019–2020), Tristan Jogminas (2019–2020), Lasse Nordahl (2019–2020), Andrei Homentcovschi (2019–2020), DongCheol Jwa (2019), HyeEun Song (2019), MyoungHoon Han (2019), Hyunji Lee (2019), Jingyu Chen (2019), Rayan Al Atab (2019), Cynthia Wang (2019), Rohit Sriram (2019), Jorge Carlos (2019), Owen Yang (2017–2019), Joshua Cao (2017–2019), Kyle Norland (2018), Fangfang Fu (2017), Angel Castro (2017), Rushabh Shah (2016–2019), Karthik Gajulapalli (2016–2018).
- *University of Zaragoza*: Ismael Rodriguez* (2015–2016).

Past high school students: Ayush Mishra (2025–2026), Samir Adhikari (2025–2026), Sitara Chakkamadathil (2024–2025), Enyu Ling (2023–2024), Eddie Wu (2023–2024), Thien-Andy Le (2023–2024), Akhil Shastry (2023–2024) .

PUBLICATIONS

Journal Articles

- [J-18] Ikechukwu Oranekwu, Lavanya Elluri, **Roberto Yus**, and Anantaa Kotal. “Scalable automation for IoT cyberSecurity compliance: Ontology-driven reasoning for real-time assessment”. In: *Comput. Secur.* 161 (2026), p. 104711. *IF: 5.4 JCR2025; Google h5-index 112*.
- [J-17] Jun Ma, Dimitrije Panic, **Roberto Yus**, and Georgios Bouloukakis. “A customizable benchmarking tool for evaluating personalized thermal comfort provisioning in smart spaces using Digital Twins”. In: *Pervasive Mob. Comput.* 112 (2025), p. 102076. *IF: 3.5 JCR2025, Q1; Google h5-index 38*.
- [J-16] Aamir Hamid, Hemanth Reddy Samidi, Tim Finin, Primal Pappachan, and **Roberto Yus**. “GENAIPABENCH: A Benchmark for Generative AI-based Privacy Assistants”. In: *Proc. Priv. Enhancing Technol.* 2024.3 (2024). *A CORE2023*.
- [J-15] Andrew Chio, Daokun Jiang, Peeyush Gupta, Georgios Bouloukakis, **Roberto Yus**, Sharad Mehrotra, and Nalini Venkatasubramanian. “SmartSPEC: A framework to generate customizable, semantics-based smart space datasets”. In: *Pervasive Mob. Comput.* 93 (2023). *IF: 3.5 JCR2025, Q1; Google h5-index 38*.

- [J-14] Dave Archer, Michael A August, Georgios Bouloukakis, Christopher Davison, Mamadou H Diallo, Dhrubajyoti Ghosh, Christopher T Graves, Michael Hay, Xi He, Peeter Laud, Steve Lu, Ashwin Machanavajjhala, Sharad Mehrotra, Gerome Miklau, Alisa Pankova, Shantanu Sharma, Nalini Venkatasubramanian, Guoxi Wang, and **Roberto Yus**. "Transitioning from testbeds to ships: an experience study in deploying the TIPPERS Internet of Things platform to the US Navy". In: *The Journal of Defense Modeling and Simulation* 19.3 (2022), pp. 501–517.
- [J-13] Dhrubajyoti Ghosh, Peeyush Gupta, Sharad Mehrotra, **Roberto Yus**, and Yasser Altowim. "JENNER: Just-in-time Enrichment in Query Processing". In: *Proc. VLDB Endow.* 15.11 (2022), pp. 2666–2678. *A* CORE2023* ; *Google h5-index* 73.
- [J-12] **Roberto Yus**, Georgios Bouloukakis, Sharad Mehrotra, and Nalini Venkatasubramanian. "The SemIoTic Ecosystem: A Semantic Bridge between IoT Devices and Smart Spaces". In: *ACM Trans. Internet Techn.* 22.3 (2022), 76:1–76:33. *IF: 3.135 JCR2021*; *Google h5-index* 33.
- [J-11] Peeyush Gupta, Michael J. Carey, Sharad Mehrotra, and **Roberto Yus**. "SmartBench: A Benchmark For Data Management In Smart Spaces". In: *Proc. VLDB Endow.* 13.11 (2020), pp. 1807–1820. *A* CORE2023* ; *Google h5-index* 73.
- [J-10] Yiming Lin, Daokun Jiang, **Roberto Yus**, Georgios Bouloukakis, Andrew Chio, Sharad Mehrotra, and Nalini Venkatasubramanian. "LOCATER: Cleaning WiFi Connectivity Datasets for Semantic Localization". In: *Proc. VLDB Endow.* 14.3 (2020), pp. 329–341. *A* CORE2023* ; *Google h5-index* 73.
- [J-9] Primal Pappachan, **Roberto Yus**, Sharad Mehrotra, and Johann-Christoph Freytag. "Sieve: A Middleware Approach to Scalable Access Control for Database Management Systems". In: *Proc. VLDB Endow.* 13.11 (2020), pp. 2424–2437. *A* CORE2023* ; *Google h5-index* 73.
- [J-8] **Roberto Yus**, Carlos Bobed, and Eduardo Mena. "A Knowledge-Based Approach to Enhance Provision of Location-Based Services in Wireless Environments". In: *IEEE Access* 8 (2020), pp. 80030–80048. *IF: 3.745 JCR2019*, *Q1 Computer Science*, *Information Systems*, *Q1 Engineering*, *Electrical & Electronic*, *Q2 Telecommunications*.
- [J-7] Abhay L. Kashyap, Lushan Han, **Roberto Yus**, Jennifer Sleeman, Taneeya Satyapanich, Sunil Gandhi, and Tim Finin. "Robust semantic text similarity using LSA, machine learning, and linguistic resources". In: *Lang. Resour. Evaluation* 50.1 (2016), pp. 125–161. *IF: 0.738 JCR2016*, *Q4 Computer Science*, *Interdisciplinary Applications*; *Google h5-index* 22.
- [J-6] Carlos Bobed, **Roberto Yus**, Fernando Bobillo, and Eduardo Mena. "Semantic reasoning on mobile devices: Do Androids dream of efficient reasoners?" In: *J. Web Semant.* 35 (2015), pp. 167–183. *IF: 2.55 JCR2014*, *Q1 Computer Science*, *Artificial Intelligence*, *Q1 Computer Science*, *Information Systems*, *Q1 Computer Science*, *Software Engineering*; *Google h5-index* 36.
- [J-5] **Roberto Yus**, Sergio Ilarri, and Eduardo Mena. "Real-time selection of video streams for live TV broadcasting based on Query-by-Example using a 3D model". In: *Multim. Tools Appl.* 74.8 (2015), pp. 2659–2685. *IF: 1.346 JCR2014*, *Q2 Computer Science*, *Information Systems*, *Q2 Computer Science*, *Software Engineering*, *Q2 Computer Science*, *Theory & Methods*, *Q2 Engineering*, *Electrical & Electronic*; *Google h5-index* 33.
- [J-4] **Roberto Yus**, Eduardo Mena, Sergio Ilarri, Arantza Illarramendi, and Jorge Bernad. "MultiCAMBA: a system for selecting camera views in live broadcasting of sport events using a dynamic 3D model". In: *Multim. Tools Appl.* 74.11 (2015), pp. 4059–4090. *IF: 1.346 JCR2014*, *Q2 Computer Science*, *Information Systems*, *Q2 Computer Science*, *Software Engineering*, *Q2 Computer Science*, *Theory & Methods*, *Q2 Engineering*, *Electrical & Electronic*; *Google h5-index* 33.
- [J-3] **Roberto Yus**, Eduardo Mena, Sergio Ilarri, and Arantza Illarramendi. "Flexible Access to Services in Smart Cities: Let SHERLOCK Advise Modern Citizens". In: *ERCIM News* 2014.98 (2014). URL: <http://ercim-news.ercim.eu/en98/special/flexible-access-to-services-in-smart-cities-let-sherlock-advise-modern-citizens>.
- [J-2] **Roberto Yus**, Eduardo Mena, Sergio Ilarri, and Arantza Illarramendi. "SHERLOCK: Semantic management of Location-Based Services in wireless environments". In: *Pervasive Mob. Comput.* 15 (2014), pp. 87–99. *IF: 3.5 JCR2025*, *Q1*; *Google h5-index* 38.
- [J-1] Sergio Ilarri, Eduardo Mena, Arantza Illarramendi, **Roberto Yus**, Maider Laka, and Gorka Marcos. "A friendly location-aware system to facilitate the work of technical directors when broadcasting sport

events". In: *Mob. Inf. Syst.* 8.1 (2012), pp. 17–43. *IF: 1.789 JCR2013, Q1 Computer Science, Information Systems, Q1 Telecommunications; Google h5-index 14.*

Conference Proceedings

- [C-22] Christian Badolato, Nathan Samson, Houssam Hajj Hassan, Chih-Kai Huang, Georgios Bouloukakis, Primal Pappachan, and **Roberto Yus**. "PSMark: a distributed IoT benchmark for publish/subscribe under domain-based workloads". In: *PerCom 2026-24th IEEE International Conference on Pervasive Computing and Communications*. 2026. *A* CORE2023*.
- [C-21] Christian Badolato, Kaur Kullman, Nikolaos Papadakis, Manav Bhatt, Georgios Bouloukakis, Don Engel, and **Roberto Yus**. "Your smart home exchanged 3M messages: defining and analyzing smart device passive mode". In: *23rd International Conference on Pervasive Computing and Communications (PerCom)*. 2025. (**Best Artifact Award**), *A* CORE2023*.
- [C-20] Malik Luti, David Feldman, ZhengHui Hu, Sharad Mehrotra, Marina Mendoza, Nalini Venkatasubramanian, **Roberto Yus**, and Ronald Eguchi. "FOVDA: A Federated Architecture for Overcoming Data Silos in Water Domain [Vision]". In: *Advances in Information and Communication (FICC)*. 2025.
- [C-19] Jun Ma, **Roberto Yus**, and Georgios Bouloukakis. "DigiGuide: A DT-based Occupant Guiding System for Optimizing Comfort and Energy Consumption". In: *11th IEEE International Conference on Smart Computing (SmartComp)*. 2025.
- [C-18] Nikolaos Papadakis, Athanasios N. Tsirikas, Fotios Fouskas, Krish Chatterjie, Christian Badolato, Gregory Blanc, **Roberto Yus**, Kostas Magoutis, and Georgios Bouloukakis. "A Context-Aware Benchmarking Approach for Scenario-Based Evaluation of Smart Devices". In: *15th International Conference on the Internet of Things, IOT*. Ed. by Stefan Nastic, Florian Michahelles, Sashko Ristov, Patrizio Dazzi, and Florian Wolling. ACM, 2025, pp. 122–130.
- [C-17] **Roberto Yus**, Nada Lahjouji, Georgios Bouloukakis, Sharad Mehrotra, and Nalini Venkatasubramanian. "Modeling Inhabited Smart Spaces to Support Interoperable IoT-Based Applications". In: *26th IEEE International Conference on Mobile Data Management, MDM*. IEEE, 2025, pp. 43–54. (**Nominated for Best Paper Award**); *B CORE2023*.
- [C-16] Habiba Farrukh, Nada Lahjouji, Sharad Mehrotra, Faisal Nawab, Julie Rousseau, Shantanu Sharma, Nalini Venkatasubramanian, and **Roberto Yus**. "PrivacySphere: Privacy-Preserving Smart Spaces". In: *2024 IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)*. 2024.
- [C-15] Houssam Hajj Hassan, Jun Ma, Georgios Bouloukakis, **Roberto Yus**, and Ajay Kattepur. "Efficient Scheduling of Smart Building Energy Systems Using AI Planning". In: *10th International Conference on ICT for Sustainability, ICT4S*. 2024, pp. 76–86.
- [C-14] Sarah Holmes, Kelly Doran, Zainab Balogun, Yoon Chung Kim, Lewis Davis II, Luke Zimmerman, **Roberto Yus**, and Tera L. Reynolds. "Perspectives on sensorization in the nursing home among administrators, staff, residents and their family members". In: *Annual meeting of the Gerontological Society of America*. 2024.
- [C-13] Lewis Davis II, Zainab Balogun, Luke Zimmerman, Yoon Chung Kim, **Roberto Yus**, Kelly Doran, Sarah Holmes, and Tera L. Reynolds. "Co-designing Sensorization Plans in the Nursing Home Setting". In: *American Medical Informatics Association Annual Symposium (AMIA)*. 2024.
- [C-12] Jun Ma, Dimitrije Panic, **Roberto Yus**, and Georgios Bouloukakis. "Co-zyBench: Using Co-Simulation and Digital Twins to Benchmark Thermal Comfort Provision in Smart Buildings". In: *IEEE International Conference on Pervasive Computing and Communications, PerCom 2024*. IEEE, 2024. *A* CORE2023*.
- [C-11] Satoshi Kai, Takao Kondo, Naghmeh Karimi, Konstantinos Mersinas, Marc Sel, **Roberto Yus**, and Satoru Tezuka. "International Mutual Recognition: A Description of Trust Services in US, UK, EU and JP and the Testbed "Hakoniwa"". In: *20th International Conference on Security and Cryptography, SECRYPT*. 2023, pp. 764–771. *C CORE2023*.

- [C-10] Andrew Chio, Daokun Jiang, Peeyush Gupta, Georgios Bouloukakis, **Roberto Yus**, Sharad Mehrotra, and Nalini Venkatasubramanian. “SmartSPEC: Customizable Smart Space Datasets via Event-driven Simulations”. In: *IEEE International Conference on Pervasive Computing and Communications, PerCom*. IEEE, 2022, pp. 152–162. (**Mark Weiser Best Paper Award**), *A* CORE2023*.
- [C-9] Maksim Ekin Eren, Manish Bhattarai, Nicholas Solovyev, Luke E. Richards, **Roberto Yus**, Charles Nicholas, and Boian S. Alexandrov. “One-Shot Federated Group Collaborative Filtering”. In: *21st IEEE International Conference on Machine Learning and Applications, ICMLA*. IEEE, 2022, pp. 647–652. *C CORE2023*.
- [C-8] Peeyush Gupta, Sharad Mehrotra, Shantanu Sharma, **Roberto Yus**, and Nalini Venkatasubramanian. “Sentauro: Sensor Observable Data Model for Smart Spaces”. In: *31st ACM International Conference on Information & Knowledge Management, CIKM*. ACM, 2022, pp. 3131–3140. *A CORE2023*.
- [C-7] **Roberto Yus**, Georgios Bouloukakis, Sharad Mehrotra, and Nalini Venkatasubramanian. “Abstracting Interactions with IoT Devices Towards a Semantic Vision of Smart Spaces”. In: *6th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, BuildSys*. ACM, 2019, pp. 91–100. *A CORE2018 ; Google h5-index 22*.
- [C-6] Mamadou H. Diallo, Nisha Panwar, **Roberto Yus**, and Sharad Mehrotra. “Trustworthy Privacy Policy Translation in Untrusted IoT Environments”. In: *3rd International Conference on Internet of Things, Big Data and Security, IoTBDS*. 2018, pp. 132–143.
- [C-5] Eun-Jeong Shin, **Roberto Yus**, Sharad Mehrotra, and Nalini Venkatasubramanian. “Exploring fairness in participatory thermal comfort control in smart buildings”. In: *4th ACM International Conference on Systems for Energy-Efficient Built Environments, BuildSys*. ACM, 2017, 19:1–19:10. *A CORE2018 ; Google h5-index 22*.
- [C-4] **Roberto Yus** and Eduardo Mena. “Continuous Processing of Real-Time Multimedia Requests Using Semantic Techniques”. In: *13th International Conference on Advances in Mobile Computing and Multimedia, MoMM*. ACM, 2015, pp. 216–220. *B CORE2017 ; Google h5-index 11*.
- [C-3] **Roberto Yus**, Eduardo Mena, and Enrique Solano-Bes. “Generic Rules for the Discovery of Subsumption Relationships Based on Ontological Contexts”. In: *IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology, WI-IAT*. 2015, pp. 309–312. *B CORE2017 ; Google h5-index 18*.
- [C-2] Primal Pappachan, **Roberto Yus**, Anupam Joshi, and Tim Finin. “Rafiki: A semantic and collaborative approach to community health-care in underserved areas”. In: *10th IEEE International Conference on Collaborative Computing: Networking, Applications and Worksharing, CollaborateCom*. 2014, pp. 322–331. *C CORE2014 ; Google h5-index 15*.
- [C-1] **Roberto Yus**, Eduardo Mena, Jorge Bernad, Sergio Ilarri, and Arantza Illarramendi. “Location-aware system based on a dynamic 3D model to help in live broadcasting of sport events”. In: *19th International Conference on Multimedia, MM*. 2011, pp. 1005–1008. *A* CORE2013 ; Google h5-index 58*.

Workshop Proceedings

- [W-12] Christian Badolato, Nathan Samson, Houssam Hajj Hassan, Chih-Kai Huang, Georgios Bouloukakis, Primal Pappachan, and **Roberto Yus**. “Artifact: PSMARK: A Distributed IoT Benchmark for Publish/Subscribe Under Domain-Based Workloads”. In: *The 24th International Conference on Pervasive Computing and Communications (PerCom) Workshops 2026*. 2026. *A* CORE2023*.
- [W-11] Jun Ma, Georgios Bouloukakis, Ajay Kattapur, **Roberto Yus**, and Denis Conan. “DEMSA: a DT-enabled Middleware for Self-adaptive Smart Spaces”. In: *1st International Workshop on Middleware for Digital Twin, Midd4DT*. ACM, 2023, pp. 1–6.
- [W-10] Andrew Chio, Daokun Jiang, Peeyush Gupta, Georgios Bouloukakis, **Roberto Yus**, Sharad Mehrotra, and Nalini Venkatasubramanian. “Artifact: SmartSPEC: Customizable Smart Space Datasets via Event-driven Simulations”. In: *IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events, PerCom*. IEEE, 2022, pp. 183–184. *A* CORE2023*.

- [W-9] Primal Pappachan, Martin Degeling, **Roberto Yus**, Anupam Das, Sruti Bhagavatula, William Melicher, Pardis Emami Naeini, Shikun Zhang, Lujo Bauer, Alfred Kobsa, Sharad Mehrotra, Norman M. Sadeh, and Nalini Venkatasubramanian. "Towards Privacy-Aware Smart Buildings: Capturing, Communicating, and Enforcing Privacy Policies and Preferences". In: *37th IEEE International Conference on Distributed Computing Systems Workshops, ICDCS Workshops*. 2017, pp. 193–198.
- [W-8] Primal Pappachan, **Roberto Yus**, Prajit Kumar Das, Sharad Mehrotra, Tim Finin, and Anupam Joshi. "Building a Mobile Applications Knowledge Base for the Linked Data Cloud". In: *1st International Workshop on Mobile Deployment of Semantic Technologies (MoDeST 2015) co-located with 14th International Semantic Web Conference (ISWC)*. Vol. 1506. CEUR Workshop Proceedings. 2015, pp. 14–25.
- [W-7] **Roberto Yus**, Fernando Bobillo, Carlos Bobed, and Eduardo Mena. "The OWL Reasoner Evaluation Goes Mobile". In: *4th International Workshop on OWL Reasoner Evaluation (ORE-2015) co-located with the 28th International Workshop on Description Logics (DL)*. Vol. 1387. CEUR Workshop Proceedings. 2015, pp. 38–45.
- [W-6] **Roberto Yus** and Primal Pappachan. "Are Apps Going Semantic? A Systematic Review of Semantic Mobile Applications". In: *1st International Workshop on Mobile Deployment of Semantic Technologies (MoDeST 2015) co-located with 14th International Semantic Web Conference (ISWC)*. Vol. 1506. CEUR Workshop Proceedings. 2015, pp. 2–13.
- [W-5] Carlos Bobed, Fernando Bobillo, **Roberto Yus**, Guillermo Esteban, and Eduardo Mena. "Android Went Semantic: Time for Evaluation". In: *3rd International Workshop on OWL Reasoner Evaluation (ORE) co-located with the Vienna Summer of Logic (VSL)*. Vol. 1207. CEUR Workshop Proceedings. 2014, pp. 23–29.
- [W-4] Abhay L. Kashyap, Lushan Han, **Roberto Yus**, Jennifer Sleeman, Taneeya Satyapanich, Sunil Gandhi, and Tim Finin. "Meerkat Mafia: Multilingual and Cross-Level Semantic Textual Similarity Systems". In: *8th International Workshop on Semantic Evaluation, SemEval@COLING*. 2014, pp. 416–423. *Google h5-index 23*.
- [W-3] Primal Pappachan, **Roberto Yus**, Prajit Kumar Das, Tim Finin, Eduardo Mena, and Anupam Joshi. "A Semantic Context-Aware Privacy Model for FaceBlock". In: *2nd Workshop on Society, Privacy and the Semantic Web - Policy and Technology (PrivOn) co-located with the 13th International Semantic Web Conference (ISWC)*. Vol. 1316. CEUR Workshop Proceedings. 2014.
- [W-2] **Roberto Yus**, Primal Pappachan, Prajit Kumar Das, Tim Finin, Anupam Joshi, and Eduardo Mena. "Semantics for Privacy and Shared Context". In: *2nd Workshop on Society, Privacy and the Semantic Web - Policy and Technology (PrivOn) co-located with the 13th International Semantic Web Conference (ISWC)*. Vol. 1316. CEUR Workshop Proceedings. 2014.
- [W-1] **Roberto Yus**, Carlos Bobed, Guillermo Esteban, Fernando Bobillo, and Eduardo Mena. "Android goes Semantic: DL Reasoners on Smartphones". In: *2nd International Workshop on OWL Reasoner Evaluation (ORE)*. Vol. 1015. CEUR Workshop Proceedings. 2013, pp. 46–52.

Peer-Reviewed Conference Poster/Demo Abstracts

- [P/D-14] Dylan Conklin, Primal Pappachan, and **Roberto Yus**. "BL(u)E CRAB: A User-Centric Framework for Identifying Suspicious Bluetooth Trackers". In: *IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events, PerCom*. IEEE, 2025, pp. 570–572. *A* CORE2023*.
- [P/D-13] Jun Ma, Aziz Boubaker, Sabrine Azaiez, **Roberto Yus**, and Georgios Bouloukakakis. "DEMO: Web-CozyBench - A Web-Based Platform to Benchmark Thermal Comfort Provision using Digital Twins". In: *11th IEEE International Conference on Smart Computing (SmartComp)*. 2025.
- [P/D-12] Saisricharan Malkireddy, Sumedh Kane, Sourimitra Medepalli, Satvik Racharla, Bharg Barot, Christian Badolato, and **Roberto Yus**. "LOADS: LiDAR-based Privacy-Preserving Queue Monitoring and Analysis". In: *IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events, PerCom*. IEEE, 2025, pp. 573–575. *A* CORE2023*.

- [P/D-11] Michael August, Christopher Davison, Mamadou H. Diallo, Dhrubajyoti Ghosh, Peeyush Gupta, Christopher T. Graves, Shanshan Han, Michael Holstrom, Pramod P. Khargonekar, Megan Kline, Sharad Mehrotra, Shantanu Sharma, Nalini Venkatasubramanian, Guoxi Wang, and **Roberto Yus**. “A privacy-enabled platform for COVID-19 applications: poster abstract”. In: *18th ACM Conference on Embedded Networked Sensor Systems, SenSys*. ACM, 2020, pp. 745–746. *A* CORE2018*.
- [P/D-10] Sumaya Almanee, Georgios Bouloukakis, Daokun Jiang, Sameera Ghayyur, Dhrubajyoti Ghosh, Peeyush Gupta, Yiming Lin, Sharad Mehrotra, Primal Pappachan, Eun-Jeong Shin, Nalini Venkatasubramanian, Guoxi Wang, and **Roberto Yus**. “SemIoTic: Bridging the Semantic Gap in IoT Spaces”. In: *6th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, BuildSys*. ACM, 2019, pp. 352–353. *A CORE2018 ; Google h5-index 22*.
- [P/D-9] Sameera Ghayyur, Yan Chen, **Roberto Yus**, Ashwin Machanavajjhala, Michael Hay, Gerome Miklau, and Sharad Mehrotra. “IoT-Detective: Analyzing IoT Data Under Differential Privacy”. In: *2018 International Conference on Management of Data, SIGMOD Conference*. ACM, 2018, pp. 1725–1728. *A* CORE2018 ; Google h5-index 66*.
- [P/D-8] Primal Pappachan, **Roberto Yus**, Prajit Kumar Das, Sharad Mehrotra, Tim Finin, and Anupam Joshi. “Mobipedia: Mobile Applications Linked Data”. In: *ISWC 2015 Posters & Demonstrations Track co-located with the 14th International Semantic Web Conference (ISWC)*. Vol. 1486. CEUR Workshop Proceedings. 2015. *A CORE2014 ; Google h5-index 40*.
- [P/D-7] **Roberto Yus** and Eduardo Mena. “Cooperative Network of Mobile Agents to Remotely Process User Information Requests”. In: *IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology, WI-IAT*. 2015, pp. 227–228. *B CORE2017 ; Google h5-index 18*.
- [P/D-6] **Roberto Yus** and Eduardo Mena. “Emergency Management Using SHERLOCK”. In: *13th Annual International Conference on Mobile Systems, Applications, and Services, MobiSys*. ACM, 2015, p. 495. *B CORE2017 ; Google h5-index 49*.
- [P/D-5] **Roberto Yus** and Eduardo Mena. “Mobile Endpoints: Accessing Dynamic Information from Mobile Devices”. In: *ISWC 2015 Posters & Demonstrations Track co-located with the 14th International Semantic Web Conference (ISWC)*. Vol. 1486. CEUR Workshop Proceedings. 2015. *A CORE2014 ; Google h5-index 40*.
- [P/D-4] **Roberto Yus**, Varish Mulwad, Tim Finin, and Eduardo Mena. “Infoboxer: Using Statistical and Semantic Knowledge to Help Create Wikipedia Infoboxes”. In: *ISWC Posters & Demonstrations Track a track within the 13th International Semantic Web Conference, ISWC*. Vol. 1272. CEUR Workshop Proceedings. 2014, pp. 405–408. *A CORE2014 ; Google h5-index 40*.
- [P/D-3] **Roberto Yus**, Primal Pappachan, Prajit Kumar Das, Eduardo Mena, Anupam Joshi, and Tim Finin. “FaceBlock: privacy-aware pictures for Google glass”. In: *12th Annual International Conference on Mobile Systems, Applications, and Services, MobiSys*. 2014, p. 366. *B CORE2014 ; Google h5-index 49*.
- [P/D-2] **Roberto Yus**, Eduardo Mena, Sergio Ilarri, and Arantza Illarramendi. “SHERLOCK: a system for location-based services in wireless environments using semantics”. In: *22nd International World Wide Web Conference, WWW*. International World Wide Web Conferences Steering Committee / ACM, 2013, pp. 301–304. *A* CORE2014 ; Google h5-index 80*.
- [P/D-1] **Roberto Yus**, David Antón, Eduardo Mena, Sergio Ilarri, and Arantza Illarramendi. “DEMO MultiCAMBA: A System to Assist in the Broadcasting of Sport Events”. In: *8th International ICST Conference Mobile and Ubiquitous Systems: Computing, Networking, and Services, MobiQuitous*. Vol. 104. 2011, pp. 238–242. *A CORE2013 ; Google h5-index 12*.

Books and Chapters

- [B-1] Carlos Bobed, **Roberto Yus**, Fernando Bobillo, Sergio Ilarri, Jorge Bernad, Eduardo Mena, Raquel Trillo-Lado, and Angel Luis Garrido. “Chapter 4: Emerging Semantic-Based Applications”. In: *Semantic Web: Implications for Technologies and Business Practices*. Ed. by Michael Workman. Springer International Publishing, Aug. 2015, pp. 39–83.

Dissertation

- [D-1] **Roberto Yus.** “Semantic Management of Location-Based Services in Wireless Environments”. PhD thesis. University of Zaragoza, Mar. 2016. URL: <http://ra.cps.unizar.es:8080/PUBLICATIONS/attachedFiles/document/thesis-sherlock.pdf>.

INVITED TALKS

- “Data Privacy in a World of Interconnected Devices”, University of Zaragoza (Spain), 2025.
- “Data Privacy in a World of Interconnected Devices”, Cyber Toaster Summer School, Los Alamos National Lab (LANL), 2024.
- “Data Privacy in a World of Interconnected Devices”, Cyber Practicum, CWIT, UMBC, 2024.
- “A brief History of Data Privacy”, Cyber Toaster Summer School, Los Alamos National Lab (LANL), 2023.
- “Introducing CS Students to Privacy Through a Special Topics Course”, Social Responsibility in Technology Symposium, UMBC, 2023.
- “Fair and Sustainable Thermal Comfort Provision in Smart Buildings”, Tech for Good, CWIT, UMBC, 2023.
- “(Don’t) Mind the Gap: Bridging the Worlds of People and IoT Devices”, Data Works MD, 2022.
- “(Don’t) Mind the Gap: Bridging the Worlds of People and IoT Devices”, Texas State University, 2021.
- “(Don’t) Mind the Gap: Bridging the Worlds of People and IoT Devices”, University of Maryland Baltimore County, 2021.
- “(Don’t) Mind the Gap: Bridging the Worlds of People and IoT Devices”, Purdue University, 2021.
- “(Don’t) Mind the Gap: Bridging the Worlds of People and IoT Devices”, University of Pittsburgh, 2021.
- “Reconciling Smartness and Privacy in the Internet of Things”, Postdoc Research Symposium, University of California, Irvine, 2017.
- “Defining accurate Wikipedia Infoboxes using Statistical and Semantic Knowledge”, Edit-a-thon for the visibility of women in Aragon, Zaragoza (Spain), 2015.
- “Wearable Computing: Google Glass”, Council of Computing Majors, University of Maryland, Baltimore County, 2013.

PERSONAL BACKGROUND

- | | |
|--------------------|-------------------------------------|
| <i>Nationality</i> | Spanish. |
| <i>Languages</i> | Spanish (Fluent), English (Fluent). |